AAEE CASE REPORT #8

Tarsal tunnel syndrome is relatively rare and the diagnosis may be difficult even though the etiologies for the syndrome are multiple. The symptoms are often vague but are usually burning pain and paresthesias in the toes and soles of the feet with nocturnal exacerbations. The physical examination may elicit a sensory deficit over the cutaneous distribution of the median and/or lateral plantar nerve. There may also be tenderness over the flexor retinaculum. Electrodiagnosis, especially with the addition of the orthodromic compound nerve action potential latency technique, is essential to confirm the diagnosis. The findings may be confined to the distribution of either the medial or lateral plantar nerves and, thus, both distal latencies must be determined. Surgical decompression of the tarsal tunnel has usually proven to be helpful in those patients who do not respond to conservative treatment.

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THE TARSAL TUNNEL SYNDROME

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Tarsal tunnel syndrome (TTS) is defined as an entrapment neuropathy of the tibial nerve, or one of its branches, from its entry point under the flexor retinaculum (laciniate ligament) to the end of its lateral and medial plantar and posterior calcaneal branches. The anatomy of the tarsal tunnel region has been well described by many authors and is seen in Fig. 1.[1-5,10,13] Tarsal tunnel syndrome is relatively rare and is frequently misdiagnosed or not recognized, as the symptoms are often quite vague.[10,17,18] The term was introduced by Keck in 1962[13] and it is analogous to the carpal tunnel syndrome, although the physical findings and signs are not nearly as definite. The paresthesias and pain associated with the entrapment easily can be confused with interdigital neuroma, peripheral neuritis, tenosynovitis, plantar fasciitis, plantar calluses, acute strain of the medial longitudinal arch, and peripheral vascular disease.[3,11,13,17,20]

Many etiological factors have been proposed, mainly alluding to the relative fixed position in a nonyielding fibro-osseous tunnel in which the contents are unable to adapt to the traction change. Some of the possible etiologic factors are space-occupying lesions; tenosynovitis (especially in patients with rheumatoid arthritis); chronic thrombophlebitis with pressure on the nerve; chronic foot strain; or abnormal calcaneal eversion relative to an adducted talus and an abducted forefoot, which puts tension on the ligaments and fibrous structures of the tarsal tunnel and the posterior tibial nerve.[1,2,7,10,12,14-17,21] Systemic diseases such as gout with urate deposits can cause synovial thickening which results in a compression. Hypertrophy of the abductor hallucis muscle or an accessory muscle belly of the short digital flexors can cause compression.[12,21] Other factors listed in the literature are rapid weight gain, connective tissue changes associated with aging, fluid retention, arthritic bony spurring, and constant squatting.[2,13,21] Trauma such as ankle sprains involving the deltoid ligament, medial malleolus, and/or calcaneal fractures have also been implicated.[14] As with other entrapments, TTS is commonly associated with peripheral neuropathy.[10] The patient can have entrapment neuropathy of the tibial nerve or selective involvement of one of its branches (Fig. 1).

The symptoms include numbness and tingling of the toes and soles of the foot, followed by burning paresthesias and pain.[14] The symptoms are initially intermittent, but can be accentuated by prolonged standing or walking, and usually become more intense as the day progresses.[2,14] The symptoms are determined by the site and duration of compression.[1] Pain at night is common and often severe enough to awaken the patient.[2,6] As the pain and paresthesias become progressively more constant, and more severe, there may be proximal radiation to the calf and leg.[2,12] The pain may be

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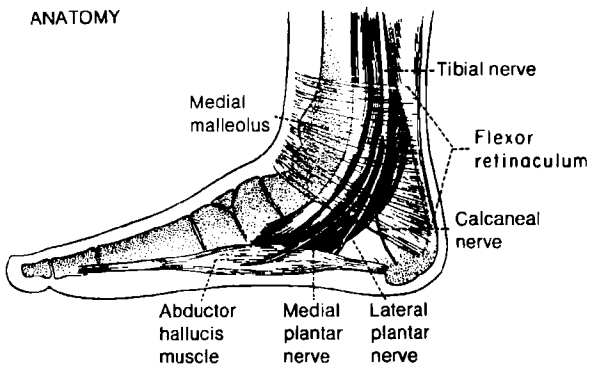


Figure 1. The tibial nerve in the region of the posterior medial aspect of the ankle dividing under the cover of the flexor retinaculum (laciniate ligament) into the calcaneal, medial, and lateral plantar nerves. (Reprinted with permission from Arch Phys Med Rehabil.[3]*)*

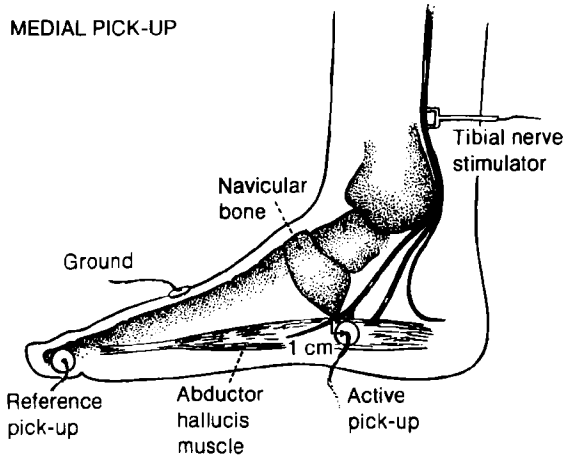


Figure. 2. The medial plantar nerve conduction latency technique. The active pickup electrode is over the motor point of the abductor hallucis muscle, 1 cm behind and below the navicular tubercle. The reference electrode is placed on the first toe. (Reprinted with permission from Arch Med Rehabil.[3]*)*

relieved by removal of the shoe, foot massage, or elevation.[2,12] Some authors feel that symptoms from acute injury, such as fractures and dislocations, are different than those from insidious compression.[14]

The following signs may be seen on examination: an alteration of sensation of the foot, especially pin prick and two point discrimination; a positive Tinel's sign over the tarsal tunnel region and flexor retinaculum; and forcing the heel into a valgus position may aggravate the symptoms, while forcing it into a varus position may cause sufficient slack in the laciniate ligament to decrease the symptoms.[2,6,14] The sensory modalities in the foot are not perceived as distinctively as those in the hand; hence, patients often have difficulty determining the area of foot pain.[14] With motor weakness, impairment of flexion of the metatarso-phalangeal joints and extension at the interphalangeal joints of all the toes may be noted. Also, it may be difficult to judge intrinsic foot weakness, especially if the foot has distorted architecture.[14] The localization of the process can be tested by injection of an anti-inflammatory agent or anesthetic agent into the tarsal tunnel to block the posterior tibial nerve.[6,14] An x-ray of the ankle region may reveal evidence of degenerative arthritis, old fractures, bone spicules, accessory ossicles, etc.[17] Electrodiagnostic studies showing prolonged sensory and/or motor distal latencies can be very useful and reliable in making the diagnosis.[1,3,7,9,11,18,19,21]

CLINICAL HISTORY

A 63-year-old white female complained of pain, burning, and numbness on the bottom of her right foot for the last few months. The pain was nearly continuous and worsened with prolonged standing and long distance walking, and at night. There was no history of trauma to the foot or ankle. She did not give a history of low back pain, diabetes mellitus or other diseases, or medication associated with a peripheral neuropathy, except a history of hypothyroidism controlled with L-thyroxine.

PHYSICAL EXAMINATION

Lower extremity strength bilaterally was within normal limits, with no atrophy of the foot muscles. She did have decreased pinprick sensation of the sole of the right foot. Tinel's sign was positive in the region of the right tarsal tunnel. Deep tendon reflexes were equal and symmetrical bilaterally.

LABORATORY

The radiologic evaluation of the right ankle revealed no bony protuberance into the tarsal tunnel region.

ELECTRODIAGNOSTIC EVALUATION

Methodology. The motor conduction studies were performed using the Johnson-Ortiz[11] technique modified by Fu et al.[3] The active electrode for the medial plantar nerve was placed over the abductor hallucis muscle, 1 cm behind and below the navicular tubercle (Fig. 2). For the lateral plantar nerve the active electrode was placed over the abductor digiti minimi at half the distance from the sole of the foot to the tip of the lateral malleolus (Fig. 3). The reference electrodes were placed over the big and little toes, respectively, and the ground electrode over the dorsum of the foot with the ankle in

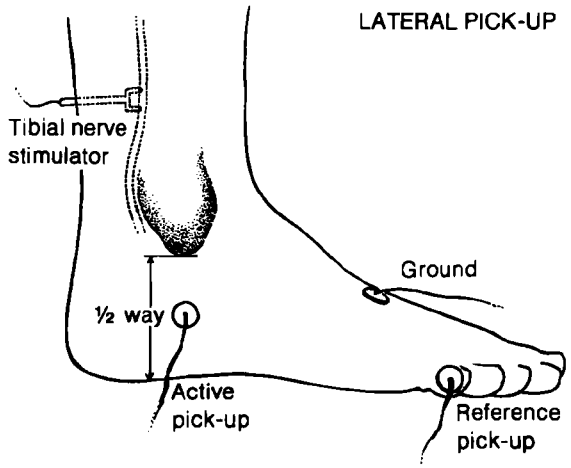


Figure 3. The lateral plantar nerve conduction latency technique. The active electrode is placed over the motor point of the abductor digiti minimi, half the distance from the sole of the foot to the tip of the lateral malleolus. The reference electrode is placed on the fifth toe. (Reprinted with permission from Arch Phys Med Rehabil.[3]*)*

the neutral position (90°). An orthodromic-evoked motor response was obtained by supramaximal stimulation of the posterior tibial nerve at the stimulation point 10 cm proximal to the recording electrode over the abductor hallucis. This stimulation point was proximal to the flexor retinaculum. The same stimulation point was used when recording from the abductor digiti minimi. The measurements were made with a flexible tape. Following these studies, compound medial and lateral plantar nerve action potentials were recorded (Figs. 6 and 7).[20] Surface skin temperature of the feet was recorded and maintained above 30°C. Needle elec-

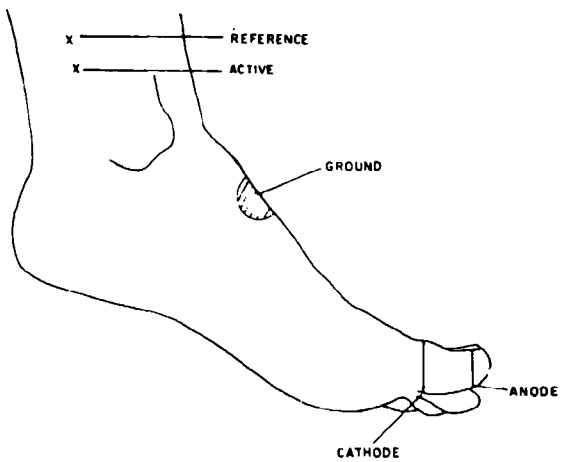


Figure 4. Medial plantar sensory nerve conduction velocity, orthodromic technique. (Reprinted with permission from DeLisa, MacKenzie: Manual of Nerve Conduction Velocity Techniques, *Raven Press, 1982.)*

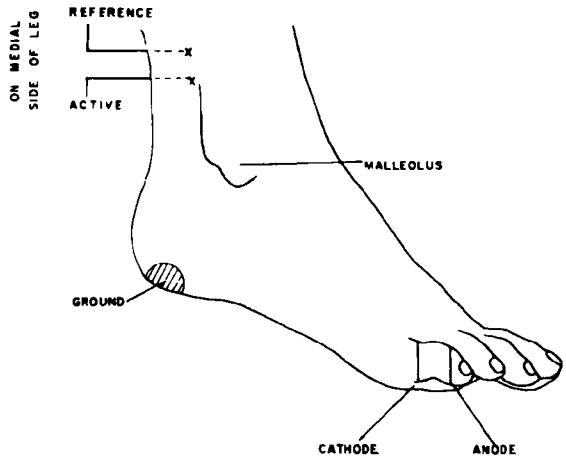


Figure 5. Lateral plantar sensory nerve conduction velocity, orthodromic technique. (Reprinted with permission from DeLisa, MacKenzie: Manual of Nerve Conduction Velocity Techniques, *Raven Press, 1982.)*

tromyography was performed using standard monopolar needle electrodes.

Results. The results of conduction parameters of the right upper and both lower extremities are noted in Table 1. Right medial and lateral plantar motor distal latencies were significantly prolonged and the evoked muscle responses were of reduced amplitude and had temporal dispersion. Compound medial and lateral plantar nerve action potentials were unobtainable. In addition, there was a slight prolongation of sural, peroneal, median, and left tibial distal latencies. Needle EMG (Table 2) demonstrated 2 to 3+ positive waves and fibrillations in the right abductor digiti minimi, abductor hallucis, and first dorsal interosseous muscles of the foot. Recruitment of the voluntary motor units was significantly reduced. There was no membrane instability present in the other muscles examined, such as the right tibialis anterior, the gastrocnemius, and the lumbosacral paraspinal muscles.

Interpretation of Clinical and Electrodiagnostic Evaluation. Both the clinical findings and the electrodiagnostic evaluation were consistent with right tarsal tunnel syndrome. Temperature correction of the sural, peroneal, and median latencies and nerve conduction velocities brought these values into the normal range.[7,8]

CLINICAL COURSE

The patient underwent surgical exploration of the right tarsal tunnel region. The flexor retinaculum

Table 1. Results of nerve conduction studies in the patient.*

Stimulate	Record	Distal latency (msec)			Conduction velocity (m/sec)			Amplitude		
		Right	Left	Normal	Right	Left	Normal	Right	Left	Normal
Posterior tibial nerve										
Medial plantar	AH	8.8	5.1	<4.8	38	45	40	1.2 mV	10 mV	>8 mV
Lateral plantar	ADM	8.5	5.7	<4.9				700 μV	8 mV	>8 mV
Medial plantar CNAP	Ankle above flexor retinaculum	NR	4.0	<3.8					10 μV	>10 μV
Lateral plantar CNAP	Ankle above flexor retinaculum	NR	4.2	<3.8					5 μV	>8 μV
Sural sensory	Behind lateral malleolus	4.6	4.6	<4.1				10 μV		>5 μV
Peroneal motor—ankle	EDB	6.4		<5.6				2 mV		>2 mV
Peroneal motor—across fibula	EDB	14.2			39		>40	2 mV		
Median sensory—wrist (14 cm)	2nd digit	4.0		<3.7				15 μV		>17 μV
Ulnar sensory—wrist (14 cm)	5th digit	3.6		<3.7				15 μV		>15 μV
Median motor— wrist	Thenar (APB)	4.5		<4.1				15 mV		>5 mV
Median motor—elbow	Thenar (APB)	9.1			52		>50	15 mV		
Ulnar motor—wrist	Hypothenar (ADM)	3.3		<4.1				15 mV		>7 mV

**Sensory conduction studies were performed antidromically and latency was measured to the peak of the sensory response.*
NR = no response, AH = abductor hallucis, ADM = abductor digiti minimi, APB = abductor pollicis brevis, EDB = extensor digitorum brevis, CNAP = compound nerve action potential.

was transected. On inspection, the posterior tibial nerve revealed that it was flattened and edematous just posterior to the medial malleolus. Postoperatively the patient noted marked improvement in her symptoms. A 3-month follow-up visit revealed no pain or burning in the right foot. She still notes slight numbness in the bottom of her foot but this is gradually improving.

ELECTROPHYSIOLOGIC EVALUATION

Discussion. Tarsal tunnel syndrome is relatively rare and the incidence is unknown.[10,17,18] This may be due to a lack of diagnostic awareness, or inaccuracy in our diagnostic techniques. The normal upper limit for conduction time for the 10-cm segment of the medial motor plantar nerve using the technique described in this case study is ≤4.8 msec (3.8 ± 0.5).[3] For the lateral plantar motor nerve it is 4.9 msec (3.9 ± 0.5).[3] This corresponds well to the 4.82 msec obtained by Oh et al.[18] (4.06 ± 0.38). However, Oh et al.[19] in reviewing their 21 cases of TTS derived from 4,000 studies in their EMG laboratory, noted that the prolonged terminal motor latency in the medial and/or lateral plantar nerve was present in only 11 cases (52.4%). Abnormal medial and lateral plantar sensory nerve conduction was observed in 19 cases (90.5%), mainly an absent potential.[19] However, their technique requires an averager as it uses 32 to 256 stimuli. The sensory potential is obtained orthodromically by placing the recording electrode above the flexor retinaculum and stimulating the big toe with a ring electrode for the medial plantar nerve, and the little toe for the lateral plantar nerve (Figs. 4 and

Table 2. Results of needle examination in the patient.

Muscle	Post insertional activity (+wave)	Spontaneous activity (fibs)	Voluntary motor unit potentials	
Right abductor hallucis	2+	2+	↓ ↓ #	↑A↑P↑D
Right first dorsal interosseous	3+	3+	↓ ↓ #	↑A↑P↑D
Right abductor digiti minimi	2+	2+	↓ ↓ #	↑A↑P↑D
Right gastrocnemius	0	0	NL	NL
Right tibialis anterior	0	0	NL	NL
Right lumbosacral paraspinals	0	0		

D = duration, # = number, A = amplitude, fibs = fibrillation potentials, P = polyphasic, + wave = positive sharp waves, NL = normal, ↑ = increased, ↓ = decreased.

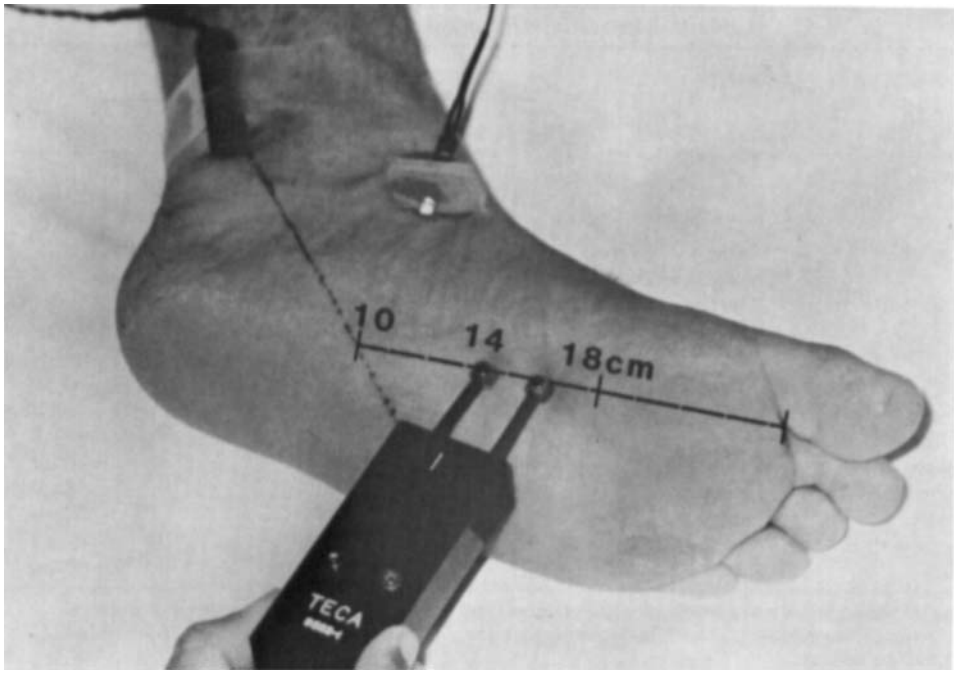


Figure 6. Orthodromic stimulation of the medial plantar nerve with the distance between the stimulating and recording electrode being 14 cm. The recording electrode must be proximal to the flexor retinaculum. (Reprinted with permission from Arch Med Phys Rehabil.[20])

5).[19] This technique is not standardized for distance, and the amplitude of the evoked nerve potentials is small: mean amplitude for the lateral plantar nerve is 1.9 μV (range 1–5 μV) and for the medial plantar nerve is 3.6 μV (range 2–6 μV).[19] The normal conduction values for this technique are 35.2 ± 3.6 m/sec for the median plantar and 31.7 ± 4.4 m/sec for the lateral plantar nerve. In our experience, these values can be difficult to obtain even in normal subjects. For those clinicians without an averager it is impossible. The 14-cm technique developed by Saeed and Gatens [20] may be very helpful in making the diagnosis. This orthodromic stimulation in the midsole of the foot for both the medial and lateral plantar nerves, provides standardized values of mixed medial plantar nerve action potential latency (3.2 ± 0.3 msec) and lateral plantar nerve action potential latency (3.2 ± 0.3 msec) using conventional equipment with or without an averager (Figs. 6 and 7).[20] These nerve action potentials are obtained by stimulating a mixed nerve, but probably represent predominantly sensory fibers and should markedly enhance our electrodiagnostic ability in TTS. If these orthodromic compound nerve action potential latencies are normal they essentially rule out TTS. However, as demonstrated in the study done by Oh et al.,[19] a normal medial and lateral plantar motor latency by itself does not necessarily rule out TTS. The sensory or mixed nerve studies must be performed.

One also has to be careful if the latency values are borderline and if there are positive sharp waves in the abductor hallucis and the abductor digiti minimi or other intrinsic foot muscles. Gatens and Saeed[4] noted positive sharp waves in 8 (11.1%) of 70 asymptomatic feet in the abductor digiti minimi and in 7 (10%) of the abductor hallucis muscles studied. Fibrillation potentials may be more of a meaningful finding but they were still present in 4 (5.7%) of the normal abductor digiti minimi muscles.[4] However, abductor hallucis and the first dorsal interosseus are two useful muscles for screening of the intrinsic foot muscles. Electrodiagnostic evaluation in a patient with possible TTS should include medial and lateral plantar motor distal latencies, sensory or mixed compound nerve studies, and the needle examination of selected intrinsic foot muscles.

Clinically, it is usually difficult to determine whether the foot symptoms are secondary to a peripheral neuropathy or an associated mechanical component at the tarsal tunnel region. A good electrophysiological evaluation will be very helpful in this differentiation. Figure 8 shows selective involvement of the left lateral plantar nerve using the compound nerve action potential technique.

In some instances, surgical decompression of

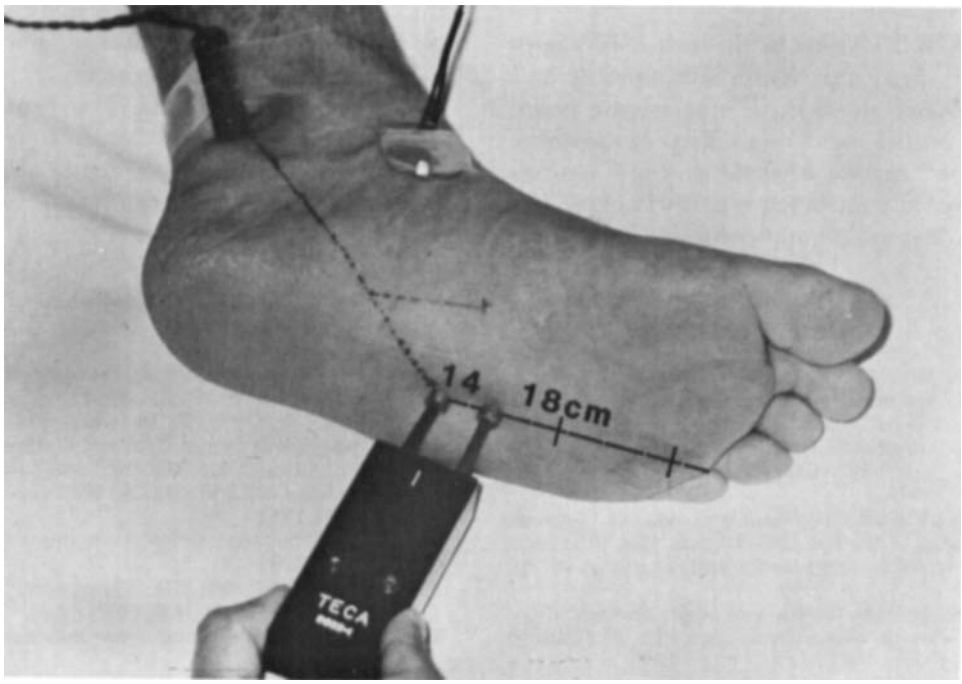


Figure 7. Orthodromic stimulation of the lateral plantar nerve with the distance between the stimulating and recording electrodes being 14 cm. The recording electrode must be proximal to the flexor retinaculum. (Reprinted with permission from Arch Med Phys Rehabil.[20])

the posterior tibial nerve and/or its lateral and medial plantar and calcaneal branches does not produce the immediate dramatic relief of symptoms that is frequently seen after carpal tunnel decompression.[14] This may be due to a coexisting peripheral neuropathy. However, surgical decompression does prevent further compromise of the nerve, and in many patients provides dramatic relief.[10,17] The literature suggests that the prognosis may be better for those whose symptoms were traumatically induced, as opposed to those with an insidious onset.[14] Technically the tarsal tunnel decompression is not that difficult, but most surgeons tend to avoid this area due to the neurovascular bundle and, thus, have less experience with this technique as opposed to the very common carpal tunnel release. A carpal tunnel release takes about 10 minutes and the tarsal tunnel takes about 20 minutes.

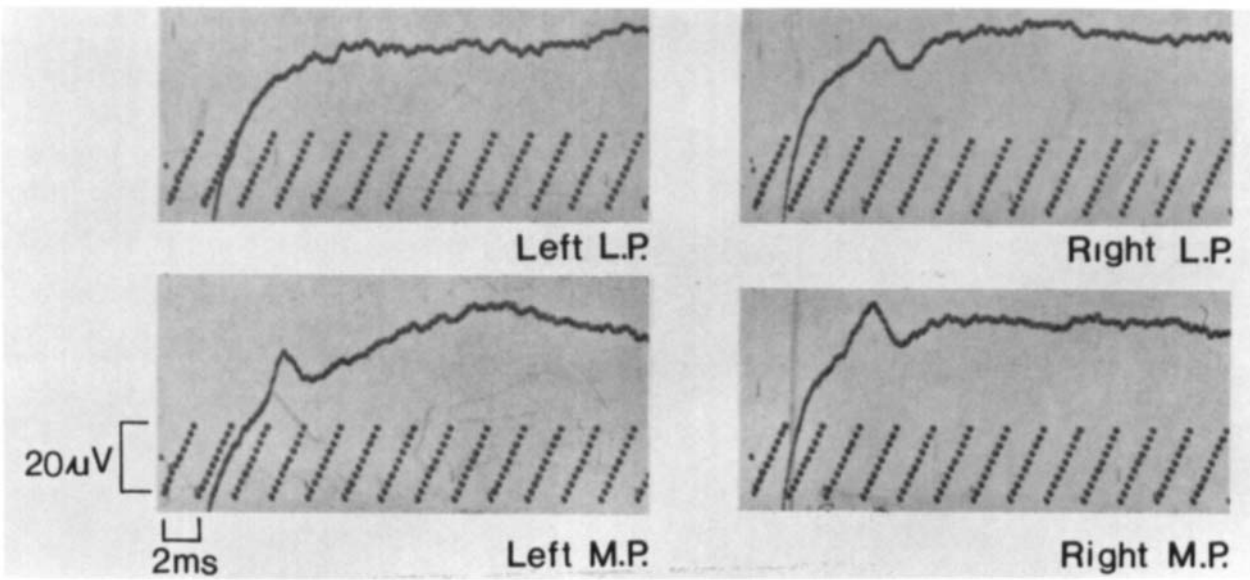


Figure 8. Absent left lateral plantar nerve action potential in a patient with a lateral plantar nerve entrapment.

SUMMARY

The diagnosis of TTS may be difficult. The symptoms are often vague and the physical findings and signs are not as definite as those noted in the analogous carpal tunnel syndrome. Electrodiagnostic testing may be essential in making the diagnosis. The addition of the averaged sensory latency with the medial and lateral plantar compound nerve action potentials should greatly increase the specificity of diagnosing this condition. The etiologies for the syndrome are multiple, even though this is a relatively rare condition. The results of surgical decompression are variable, but are usually satisfactory enough to justify this mode of therapy if the patient has not responded to conservative treatment.

REFERENCES

1. DiStefano V, Sack JT, Whittaker R, Nixon JE: Tarsal-tunnel syndrome. Review of the literature and two case reports. *Clin Orthop* 88:76–79, 1972.
2. Edwards WG, Lincoln CR, Bassett FH III, Goldner JL: The tarsal tunnel syndrome: diagnosis and treatment. *JAMA* 207:716–720, 1969.
3. Fu R, DeLisa JA, Kraft GH: Motor nerve latencies through the tarsal tunnel in normal adult subjects: standard determination corrected for temperature and distance. *Arch Phys Med Rehabil* 61:243–248, 1980.
4. Gatens PF, Saeed MA: Electromyographic findings in the intrinsic muscles of normal feet. *Arch Phys Med Rehabil* 63:317–318, 1982.
5. Goodgold J, Kopell HP, Spielholz NI: The tarsal-tunnel syndrome: objective diagnostic criteria: *N Engl J Med* 273:742–745, 1965.
6. Grabois M, Puentes J, Lidsky M: Tarsal tunnel syndrome in rheumatoid arthritis. *Arch Phys Med Rehabil* 62:401–403, 1981.
7. Halar EM, DeLisa JA, Brozovich FV: Nerve conduction velocity: relationship of skin, subcutaneous and intramuscular temperatures. *Arch Phys Med Rehabil* 61:199–203, 1980.
8. Halar EM, DeLisa JA, Brozovich FV: Peroneal nerve conduction velocity: the importance of temperature correction. *Arch Phys Med Rehabil* 62:439–443, 1981.
9. Irani KD, Grabois M, Harvey SC: Standardized technique for diagnosis of tarsal tunnel syndrome. *Am J Phys Med* 61:26–31, 1982.
10. Janecki CJ, Dovberg JL: Tarsal-tunnel syndrome caused by neurilemoma of the medial plantar nerve. A case report. *J Bone Jt Surg* 59-A:127–128, 1977.
11. Johnson EW, Ortiz PR: Electrodiagnosis of tarsal tunnel syndrome. *Arch Phys Med Rehabil* 47:776–780, 1966.
12. Kaplan PE, Kernahan WT Jr: Tarsal tunnel syndrome: an electrodiagnostic and surgical correlation. *J Bone Jt Surg* 63-A:96–99, 1981.
13. Keck C: The tarsal-tunnel syndrome. *J Bone Jt Surg* 44-A: 180–182, 1962.
14. Kuritz HM, Sokoloff TH: Tarsal tunnel syndrome. *J Am Podiatry Assoc* 65:825–840, 1975.
15. Lam SJS: A tarsal tunnel syndrome. *Lancet* 2:1354–1355, 1962.
16. Linscheid RL, Burton RC, Fredericks EJ: Tarsal-tunnel syndrome. *South Med J* 63:1313–1323, 1970.
17. Mann RA: Tarsal tunnel syndrome. *Orthop Clin North Am* 5:109–115, 1974.
18. Oh SJ, Sarala PK, Kuba T, Elmore RS: Sensory nerve conduction velocity of the plantar nerves: a superior objective diagnostic test for tarsal tunnel syndrome. *Trans Am Neurol Assoc* 103:256–258, 1978.
19. Oh SJ, Sarala PK, Kuba T, Elmore RS: Tarsal tunnel syndrome: electrophysiological study. *Ann Neurol* 5:327–330, 1979.
20. Saeed MA, Gatens PF: Compound nerve action potentials of the medial and lateral plantar nerves through the tarsal tunnel. *Arch Phys Med Rehabil* 63:304–307, 1982.
21. Willemon WK: Tarsal tunnel syndrome. A fifty year survey of the world literature and a report of two new cases. *Orthop Rev* 8:111–117, 1979.